Md Rokib Hasan Oli

2211950642

Lab Task

Q1

```
.global _start

_start:

        ldr r0,=0xff200000

        ldr r1,=0xff200050

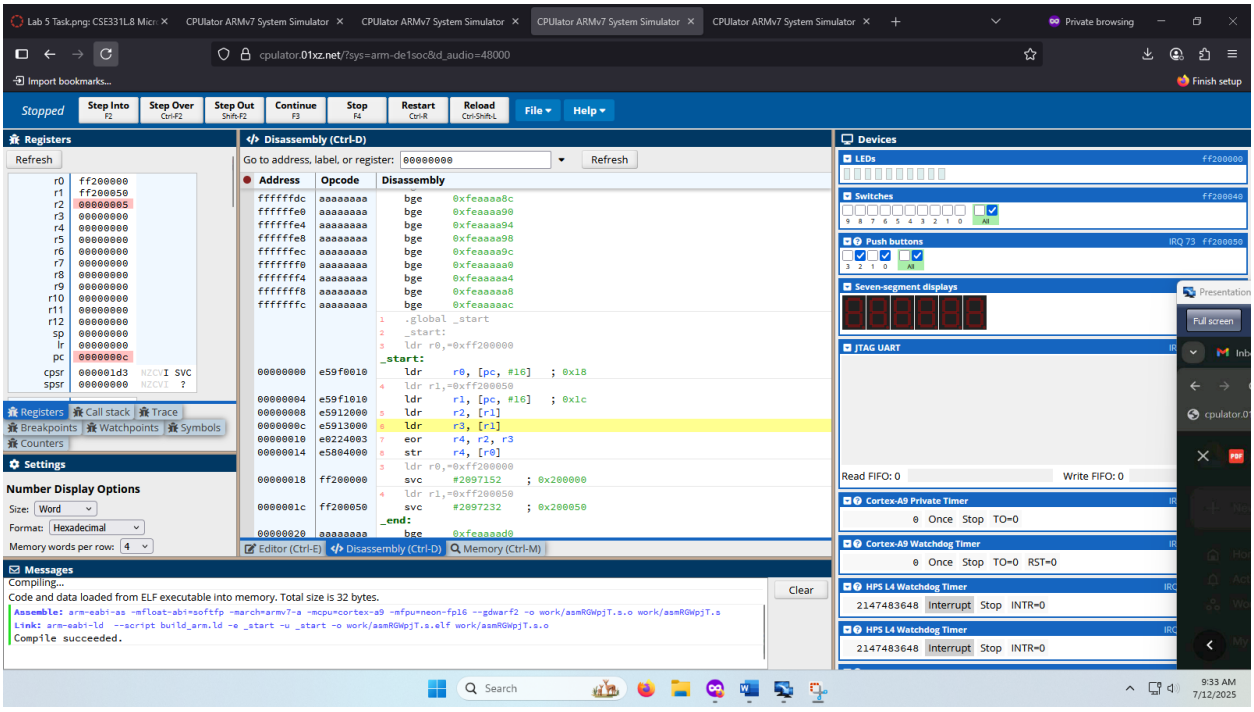
        ldr r2,[r1]

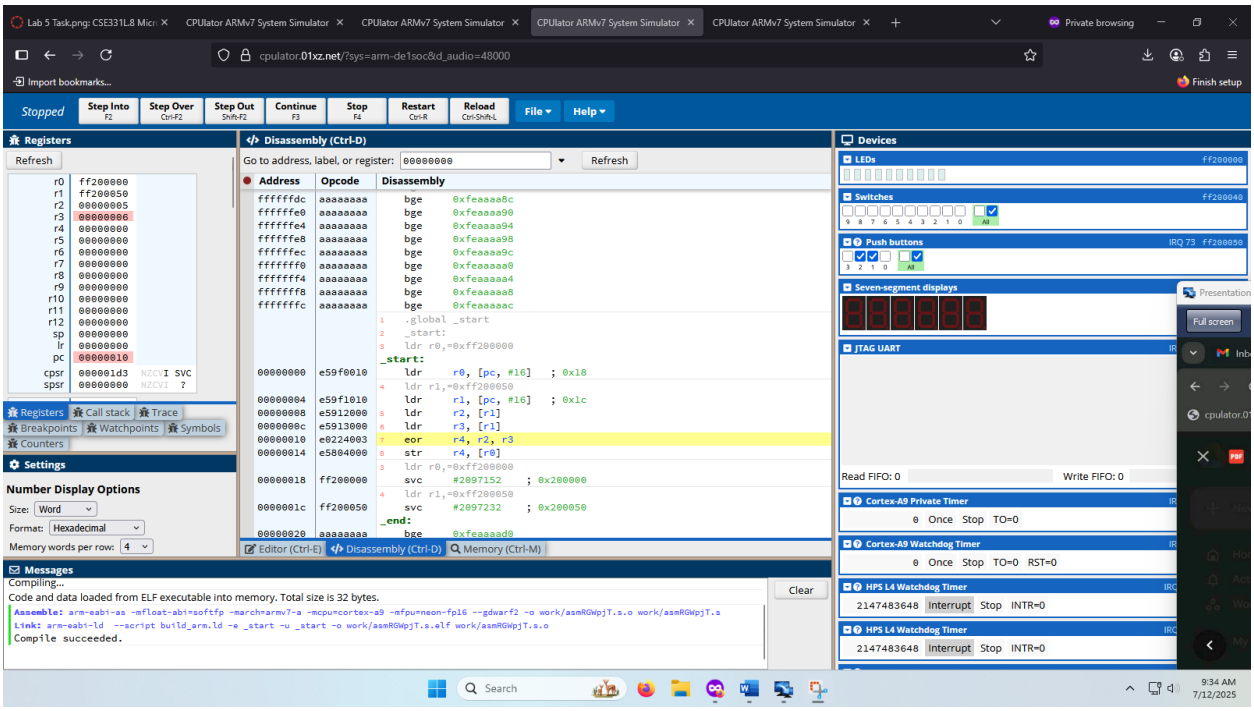
        ldr r3,[r1]

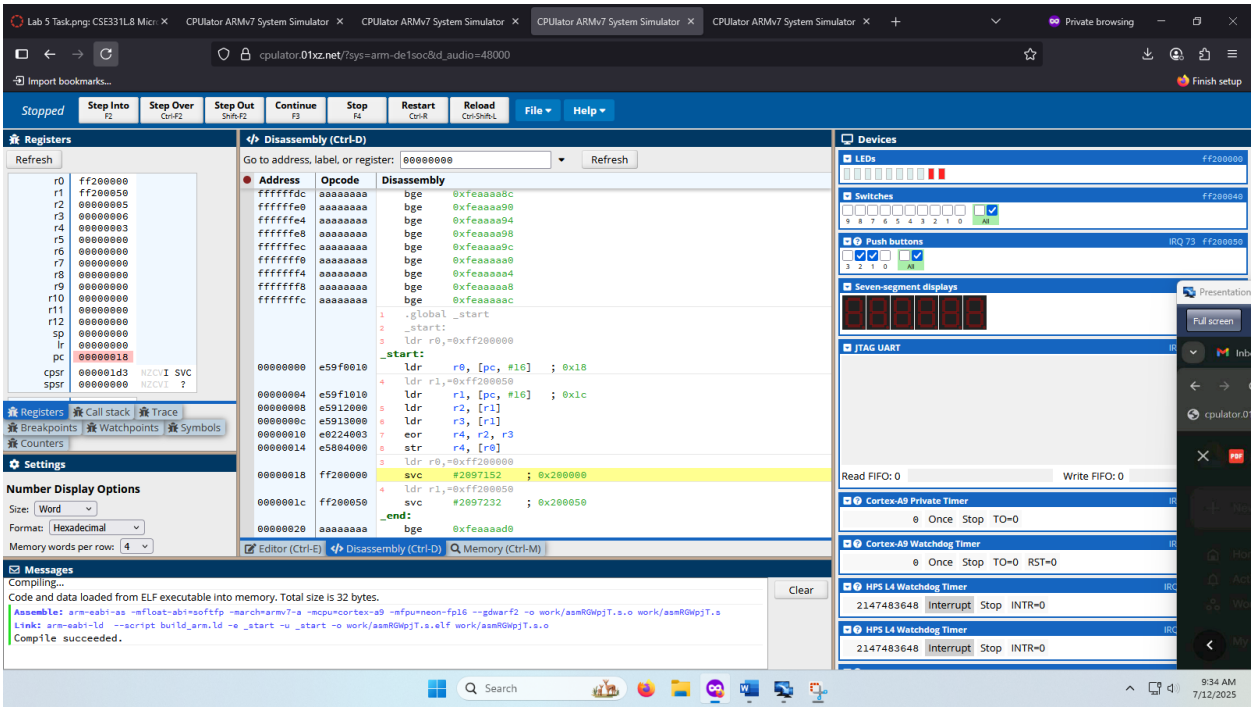
        eor r4,r2,r3

        str r4,[r0]

        .end
```




Stopped
Step Into
Step Over
Step Out
Continue
Stop
Restart
Reload
File
Help
Registers
Refresh
r0 ff200000
r1 ff200050
r2 00000005
r3 00000006
r4 00000000
r5 00000000
r6 00000000
r7 00000000
r8 00000000
r9 00000000
r10 00000000
r11 00000000
r12 00000000
sp 00000000
lr 00000000
pc 00000010
cpsr 000001d3 NZCVI SVC
spsr 00000000 NZCVI ?
Disassembly (Ctrl-D)
Go to address, label, or register: 00000000
Address Opcode Disassembly
fffffffc aaaaaaaa bge 0xfeaaaa8c
fffffffe0 aaaaaaaa bge 0xfeaaaa90
ffffffe4 aaaaaaaa bge 0xfeaaaa94
ffffffe8 aaaaaaaa bge 0xfeaaaa98
ffffffec aaaaaaaa bge 0xfeaaaa9c
fffffff0 aaaaaaaa bge 0xfeaaaaa0
fffffff4 aaaaaaaa bge 0xfeaaaaa4
fffffff8 aaaaaaaa bge 0xfeaaaaa8
fffffffc aaaaaaaa bge 0xfeaaaaac
.global _start
_start:
ldr r0,=0xff200000
_start:
00000000 e59f0010 ldr r0, [pc, #16] ; 0x18
ldr r1,=0xff200050
00000004 e59f1010 ldr r1, [pc, #16] ; 0x1c
00000008 e5912000 ldr r2, [r1]
0000000c e5913000 ldr r3, [r1]
00000010 e0224003 eor r4, r2, r3
00000014 e5804000 str r4, [r0]
ldr r0,=0xff200000
00000018 ff200000 svc #2097152 ; 0x200000
ldr r1,=0xff200050
0000001c ff200050 svc #2097232 ; 0x200050
_end:
00000020 aaaaaaaa bge 0xfeaaaad0
Editor (Ctrl-E)
Disassembly (Ctrl-D)
Memory (Ctrl-M)
Registers
Call stack
Trace
Breakpoints
Watchpoints
Symbols
Counters
Settings
Number Display Options
Size: Word
Format: Hexadecimal
Memory words per row: 4
Messages
Compiling...
Code and data loaded from ELF executable into memory. Total size is 32 bytes.
Assemble: arm-eabi-as -mfloat-abi=softfp -march=armv7-a -mcpu=cortex-a9 -mfpu=neon-fp16 --gdwarf2 -o work/asmRGWpjT.s.o work/asmRGWpjT.s
Link: arm-eabi-ld --script build_arm.ld -e _start -u _start -o work/asmRGWpjT.s.elf work/asmRGWpjT.s.o
Compile succeeded.
Clear
Devices
LEDs ff200000
Switches ff200040
Push buttons IRQ 73 ff200050
Seven-segment displays
JTAG UART
Read FIFO: 0
Write FIFO: 0
Cortex-A9 Private Timer
0 Once Stop TO=0
Cortex-A9 Watchdog Timer
0 Once Stop TO=0 RST=0
HPS L4 Watchdog Timer
2147483648 Interrupt Stop INTR=0
Presentation
Full screen
Search
9:34 AM
7/12/2025


Stopped
Step Into
Step Over
Step Out
Continue
Stop
Restart
Reload
File
Help
Registers
Refresh
r0 ff200000
r1 ff200050
r2 00000005
r3 00000006
r4 00000003
r5 00000000
r6 00000000
r7 00000000
r8 00000000
r9 00000000
r10 00000000
r11 00000000
r12 00000000
sp 00000000
lr 00000000
pc 00000018
cpsr 000001d3 NZCVI SVC
spsr 00000000 NZCVI ?
Disassembly (Ctrl-D)
Go to address, label, or register: 00000000
Address Opcode Disassembly
fffffffc aaaaaaaa bge 0xfeaaaa8c
fffffffe0 aaaaaaaa bge 0xfeaaaa90
ffffffe4 aaaaaaaa bge 0xfeaaaa94
ffffffe8 aaaaaaaa bge 0xfeaaaa98
ffffffec aaaaaaaa bge 0xfeaaaa9c
fffffff0 aaaaaaaa bge 0xfeaaaaa0
fffffff4 aaaaaaaa bge 0xfeaaaaa4
fffffff8 aaaaaaaa bge 0xfeaaaaa8
fffffffc aaaaaaaa bge 0xfeaaaaac
.global _start
_start:
ldr r0,=0xff200000
_start:
00000000 e59f0010 ldr r0, [pc, #16] ; 0x18
ldr r1,=0xff200050
00000004 e59f1010 ldr r1, [pc, #16] ; 0x1c
00000008 e5912000 ldr r2, [r1]
0000000c e5913000 ldr r3, [r1]
00000010 e0224003 eor r4, r2, r3
00000014 e5804000 str r4, [r0]
ldr r0,=0xff200000
00000018 ff200000 svc #2097152 ; 0x200000
ldr r1,=0xff200050
0000001c ff200050 svc #2097232 ; 0x200050
_end:
00000020 aaaaaaaa bge 0xfeaaaad0
Editor (Ctrl-E)
Disassembly (Ctrl-D)
Memory (Ctrl-M)
Registers
Call stack
Trace
Breakpoints
Watchpoints
Symbols
Counters
Settings
Number Display Options
Size: Word
Format: Hexadecimal
Memory words per row: 4
Messages
Compiling...
Code and data loaded from ELF executable into memory. Total size is 32 bytes.
Assemble: arm-eabi-as -mfloat-abi=softfp -march=armv7-a -mcpu=cortex-a9 -mfpu=neon-fp16 --gdwarf2 -o work/asmRGWpjT.s.o work/asmRGWpjT.s
Link: arm-eabi-ld --script build_arm.ld -e _start -u _start -o work/asmRGWpjT.s.elf work/asmRGWpjT.s.o
Compile succeeded.
Clear
Devices
LEDs ff200000
Switches ff200040
Push buttons IRQ 73 ff200050
Seven-segment displays
JTAG UART
Read FIFO: 0
Write FIFO: 0
Cortex-A9 Private Timer
0 Once Stop TO=0
Cortex-A9 Watchdog Timer
0 Once Stop TO=0 RST=0
HPS L4 Watchdog Timer
2147483648 Interrupt Stop INTR=0
Presentation
Full screen
Search
9:34 AM
7/12/2025

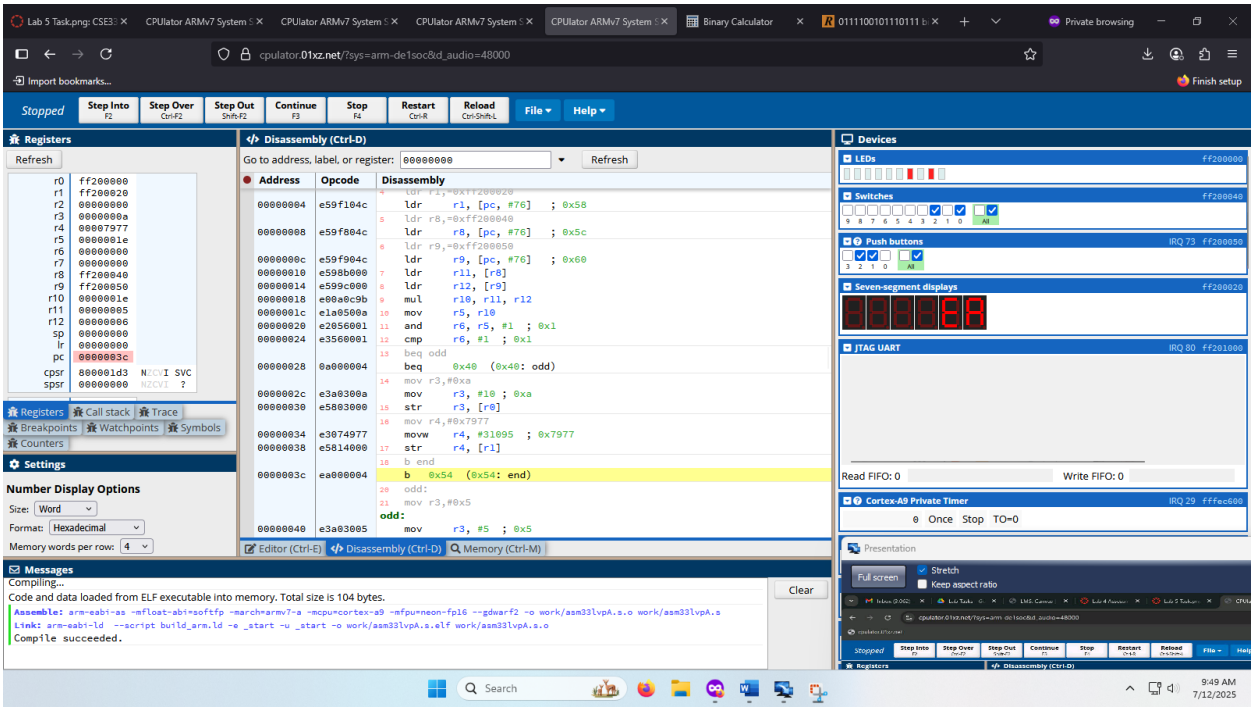
Q2

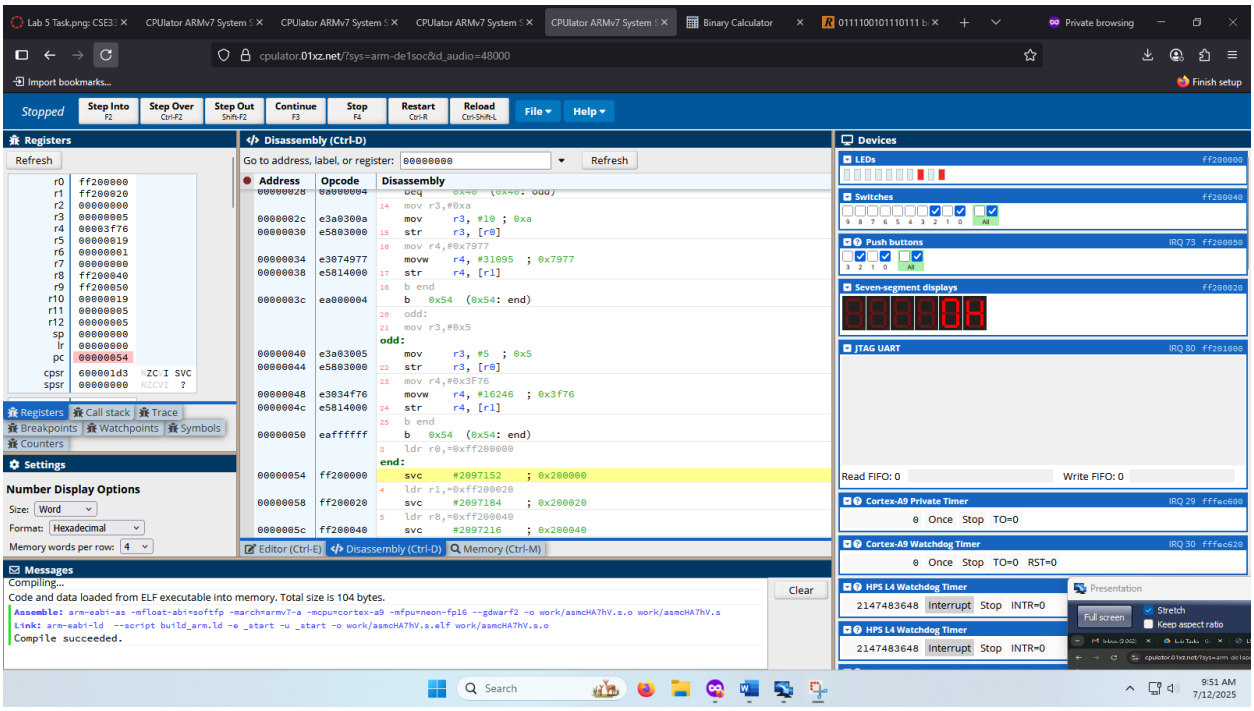
```
  .global _start
_start:
        ldr r0,=0xff200000
        ldr r1,=0xff200020
        ldr r8,=0xff200040
        ldr r9,=0xff200050
        ldr r11,[r8]
        ldr r12,[r9]
        mul r10,r11,r12
        mov r5,r10
        and r6,r5,#1
        cmp r6,#1
        beq odd
        mov r3,#0xa
        str r3,[r0]
        mov r4,#0x7977
        str r4,[r1]
        b end

        odd:
        mov r3,#0x5
        str r3,[r0]
        mov r4,#0x3F76
        str r4,[r1]
```

```
b end


end:
.end
```


Stopped
Step Into
Step Over
Step Out
Continue
Stop
Restart
Reload
File
Help
Registers
Refresh
r0 ff200000
r1 ff200020
r2 00000000
r3 0000000a
r4 00007977
r5 0000001e
r6 00000000
r7 00000000
r8 ff200040
r9 ff200050
r10 0000001e
r11 00000005
r12 00000006
sp 00000000
lr 00000000
pc 0000003c
cpsr 800001d3 NZCVI SVC
spsr 00000000 NZCVI ?
Registers
Call stack
Trace
Breakpoints
Watchpoints
Symbols
Counters
Settings
Number Display Options
Size: Word
Format: Hexadecimal
Memory words per row: 4
Disassembly (Ctrl-D)
Go to address, label, or register: 00000000
Refresh
Address Opcode Disassembly
00000084 e59f104c ldr r1, [pc, #76] ; 0x58
00000088 e59f804c ldr r8, [pc, #76] ; 0x5c
0000008c e59f904c ldr r9, [pc, #76] ; 0x60
00000010 e598b000 ldr r11, [r8]
00000014 e599c000 ldr r12, [r9]
00000018 e00a0c9b mul r10, r11, r12
0000001c e1a0500a mov r5, r10
00000020 e2056001 and r6, r5, #1 ; 0x1
00000024 e3560001 cmp r6, #1 ; 0x1
00000028 0a000004 beq 0x40 (0x40: odd)
0000002c e3a0300a mov r3, #10 ; 0xa
00000030 e5803000 str r3, [r0]
00000034 e3074977 movw r4, #31095 ; 0x7977
00000038 e5814000 str r4, [r1]
0000003c ea000004 b 0x54 (0x54: end)
00000040 e3a03005 mov r3, #5 ; 0x5
Editor (Ctrl-E)
Disassembly (Ctrl-D)
Memory (Ctrl-M)
Devices
LEDs ff200000
Switches ff200040
Push buttons IRQ 73 ff200050
Seven-segment displays ff200020
JTAG UART IRQ 80 ff201000
Read FIFO: 0
Write FIFO: 0
Cortex-A9 Private Timer IRQ 29 fffec600
0 Once Stop TO=0
Presentation
Full screen
Stretch
Keep aspect ratio
Messages
Compiling...
Code and data loaded from ELF executable into memory. Total size is 104 bytes.
Assemble: arm-eabi-as -mfloat-abi=softfp -march=armv7-a -mcpu=cortex-a9 -mfpu=neon-fp16 --gdwarf2 -o work/asm33lvpA.s.o work/asm33lvpA.s
Link: arm-eabi-ld --script build_arm.ld -e _start -u _start -o work/asm33lvpA.s.elf work/asm33lvpA.s.o
Compile succeeded.
Clear
Search
9:49 AM
7/12/2025



Class Task

Q1

```
.global _start
_start:
        mov r0,#3
        mov r1,#3
        mov r2,#5
        ldr r3,=0xff200000
        ldr r5,=0xff200020
        add r10,r0,r1
        cmp r10,r2
```

```
        bgt logic1
        b NRT

logic1:
        add r10,r1,r2
        cmp r10,r0
        bgt logic2
        b NRT
logic2:
        add r10,r0,r2
        cmp r10,r1
        b RT

NRT:
        mov r4,#0x2
        str r4,[r3]
        mov r6,#0x7C
        str r6,[r5]
        b end

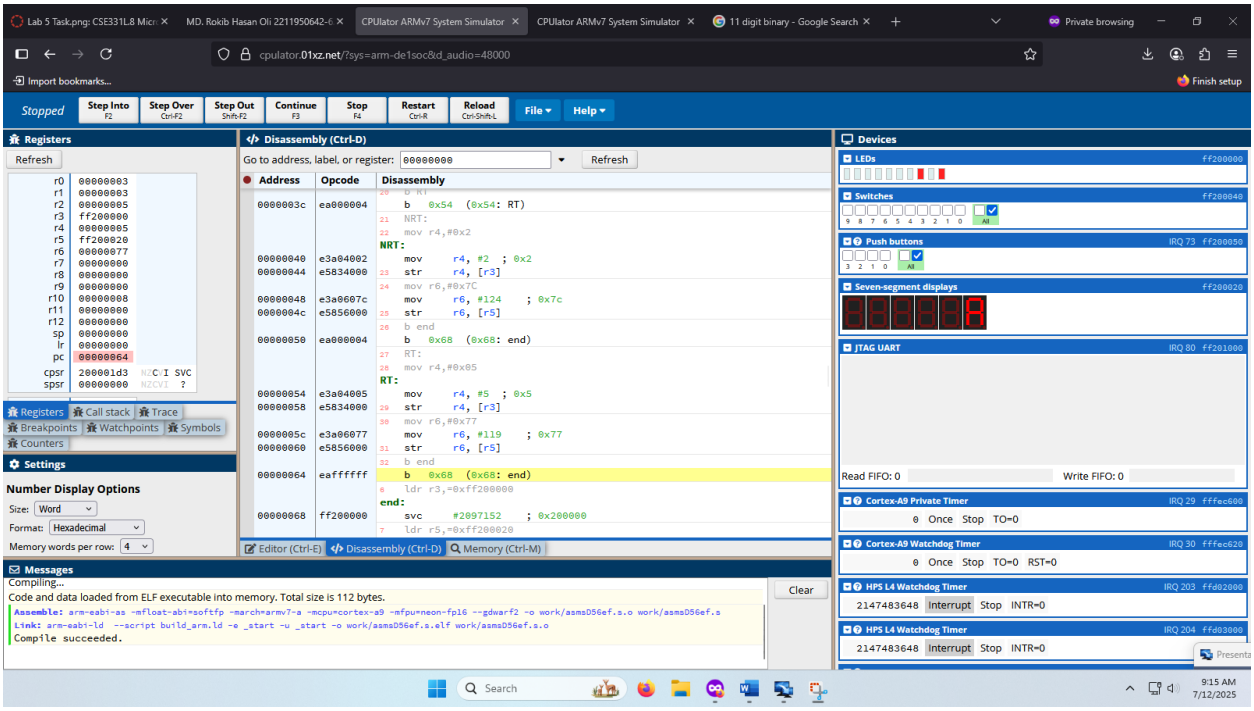
RT:
        mov r4,#0x05
        str r4,[r3]
        mov r6,#0x77
```

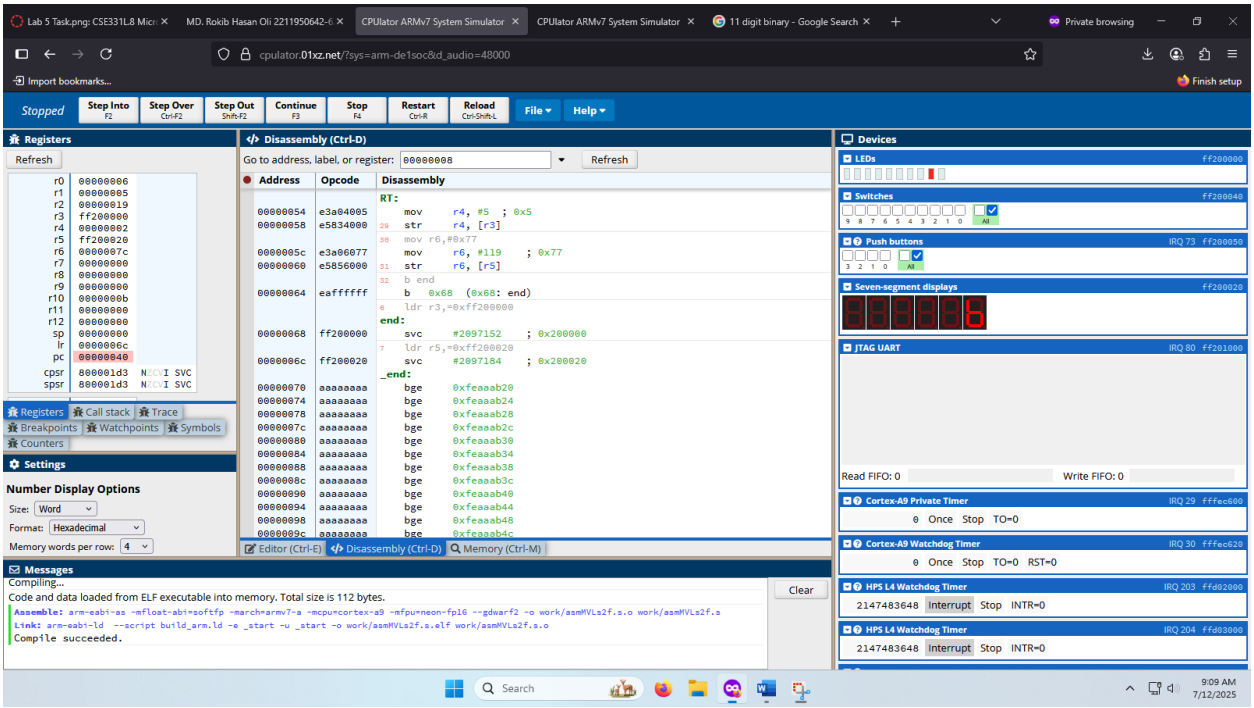
str r6,[r5]

b end

end:

.end

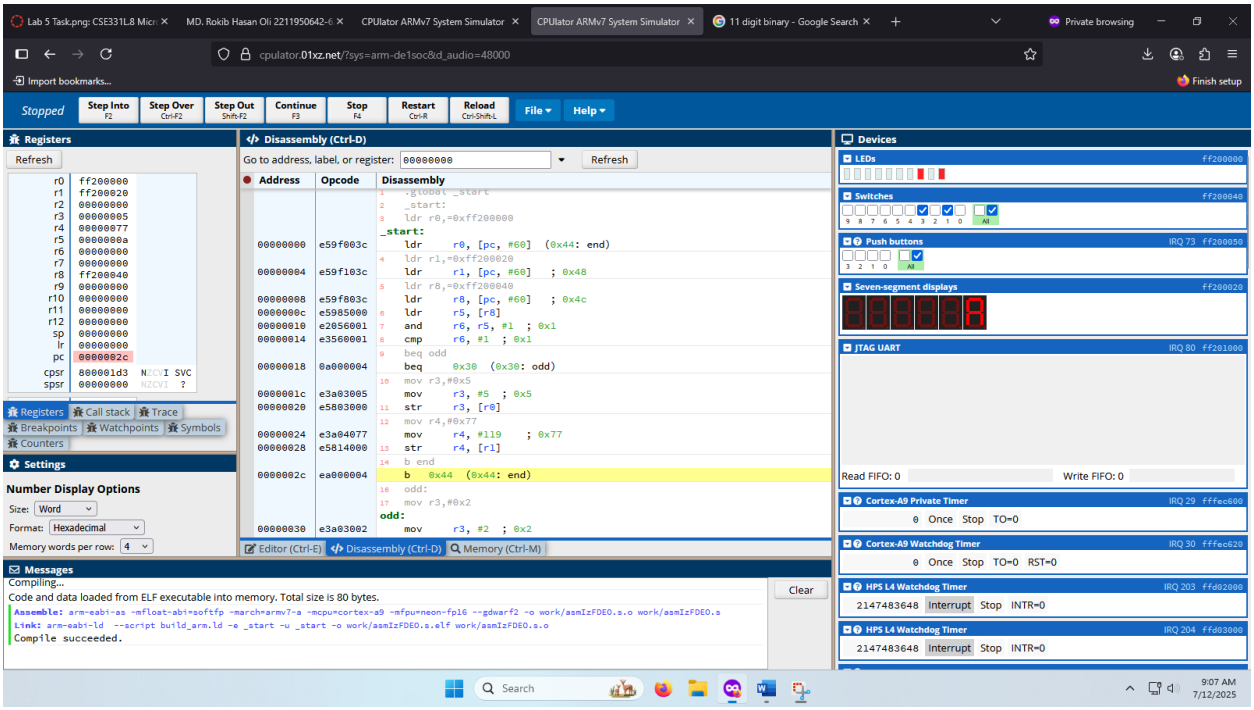


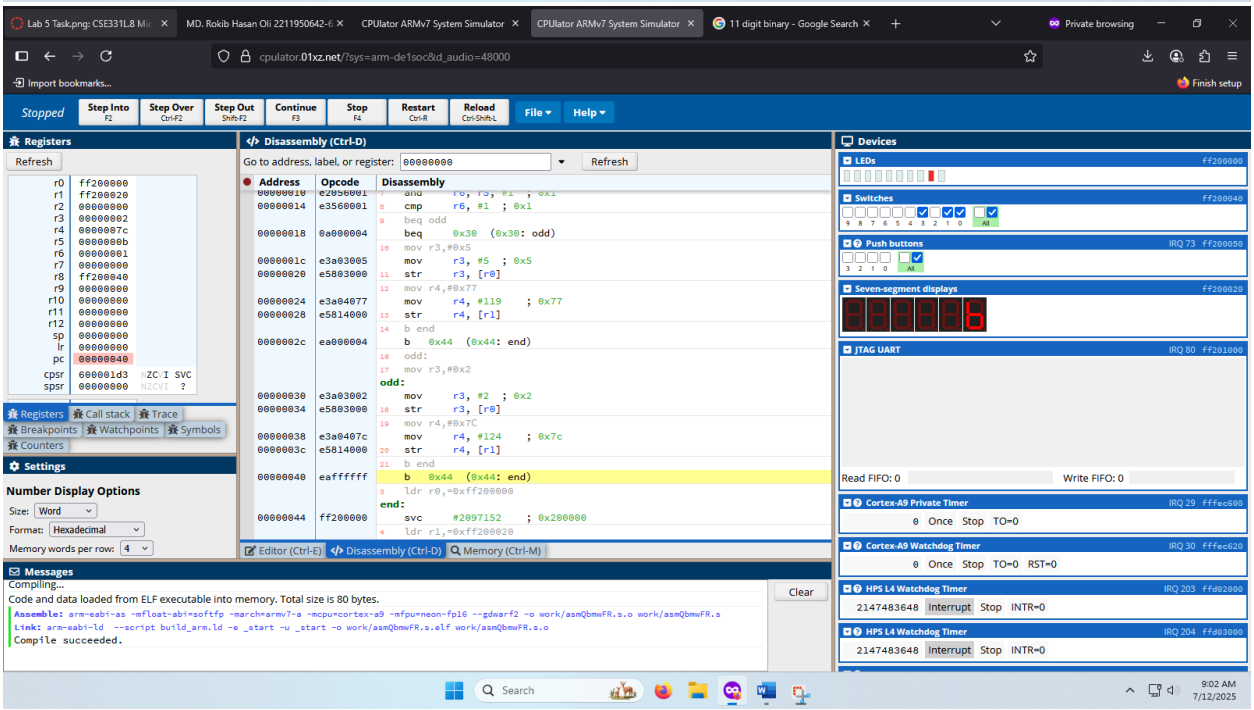


Q2

```
.global _start
_start:
        ldr r0,=0xff200000
        ldr r1,=0xff200020
        ldr r8,=0xff200040
        ldr r5,[r8]
        and r6,r5,#1
        cmp r6,#1
        beq odd
        mov r3,#0x5
        str r3,[r0]
        mov r4,#0x77
        str r4,[r1]
        b end

        odd:
        mov r3,#0x2
        str r3,[r0]
        mov r4,#0x7C
        str r4,[r1]
        b end
        end:
        .end
```





Q3

```
.global _start

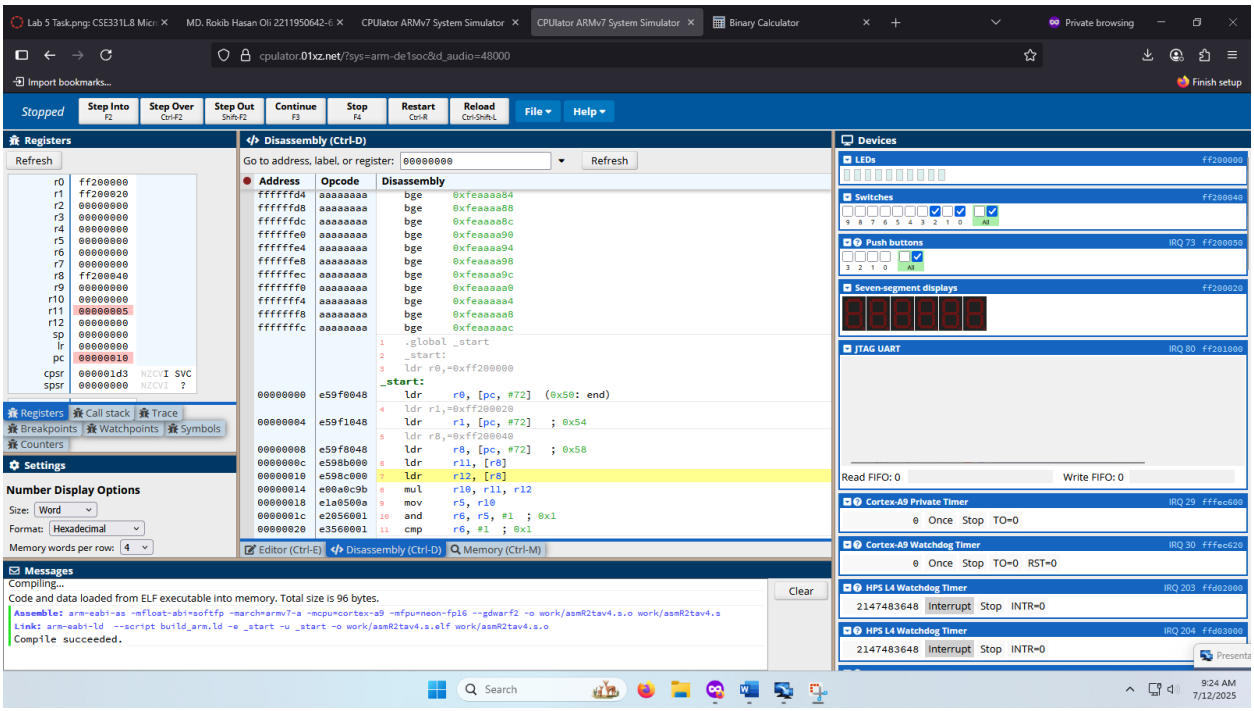
_start:

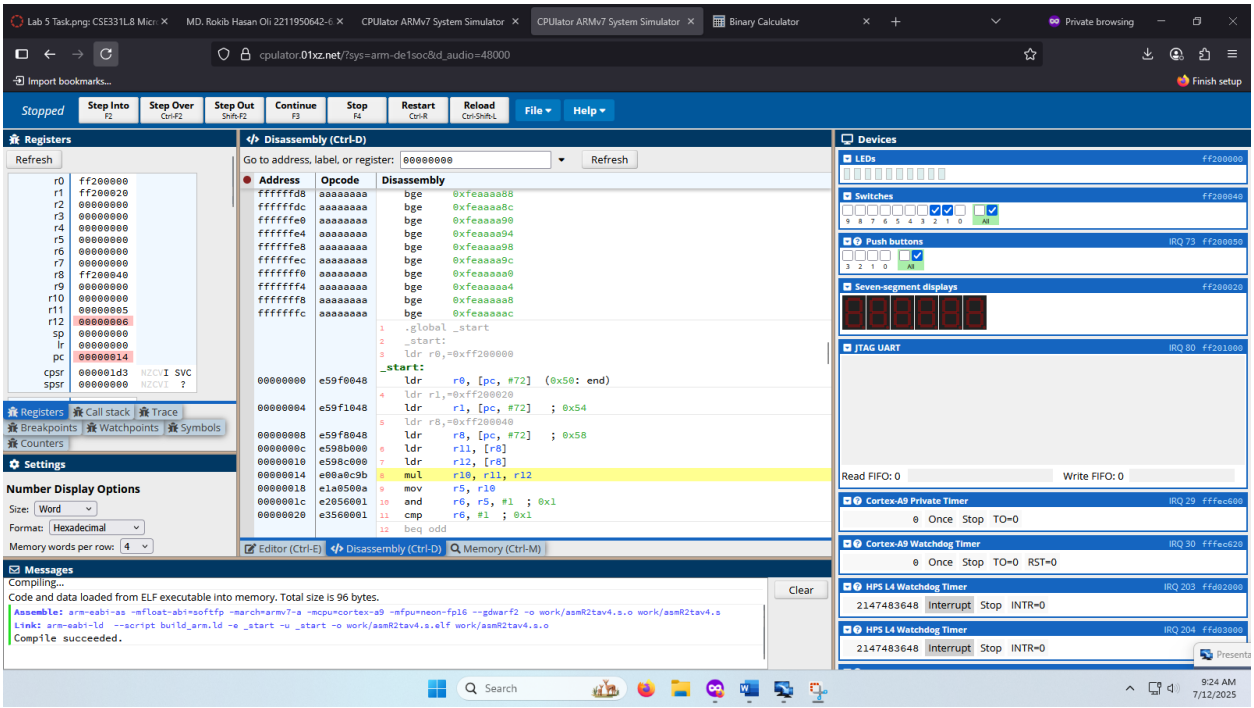
        ldr r0,=0xff200000
```

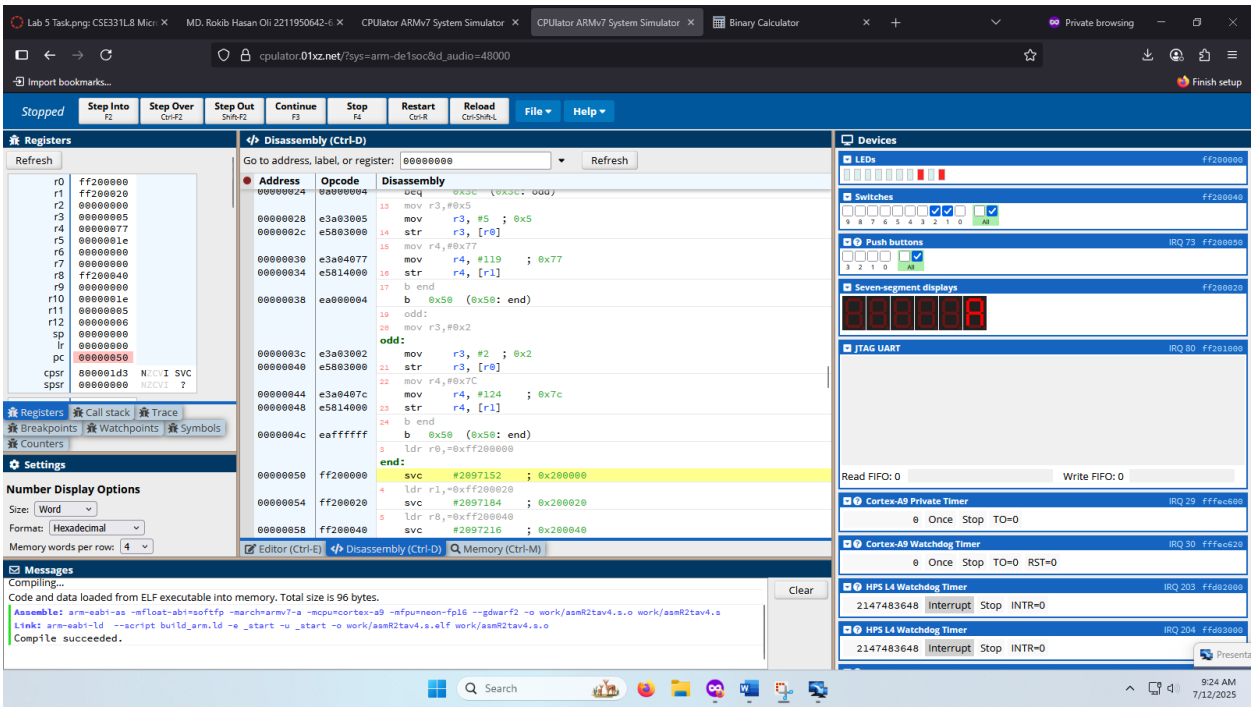
```
ldr r1,=0xff200020
        ldr r8,=0xff200040
        ldr r11,[r8]
        ldr r12,[r8]
        mul r10,r11,r12
        mov r5,r10
        and r6,r5,#1
        cmp r6,#1
        beq odd
        mov r3,#0x5
str r3,[r0]
        mov r4,#0x77
        str r4,[r1]
        b end

        odd:
        mov r3,#0x2
str r3,[r0]
        mov r4,#0x7C
        str r4,[r1]
        b end

        end:
        .end
```


Stopped
Step Into F2
Step Over Ctrl-F2
Step Out Shift-F2
Continue F3
Stop F4
Restart Ctrl-R
Reload Ctrl-Shift-L
File
Help
Registers
Refresh
r0 ff200000
r1 ff200020
r2 00000000
r3 00000000
r4 00000000
r5 00000000
r6 00000000
r7 00000000
r8 ff200040
r9 00000000
r10 00000000
r11 00000005
r12 00000000
sp 00000000
lr 00000000
pc 00000010
cpsr 000001d3 NZCVI SVC
spsr 00000000 NZCVI ?
Registers
Call stack
Trace
Breakpoints
Watchpoints
Symbols
Counters
Settings
Number Display Options
Size: Word
Format: Hexadecimal
Memory words per row: 4
Disassembly (Ctrl-D)
Go to address, label, or register: 00000000
Refresh
Address Opcode Disassembly
ffffffd4 aaaaaaaa bge 0xfeaaaa84
ffffffd8 aaaaaaaa bge 0xfeaaaa88
ffffffdc aaaaaaaa bge 0xfeaaaa8c
ffffffe0 aaaaaaaa bge 0xfeaaaa90
ffffffe4 aaaaaaaa bge 0xfeaaaa94
ffffffe8 aaaaaaaa bge 0xfeaaaa98
ffffffec aaaaaaaa bge 0xfeaaaa9c
fffffff0 aaaaaaaa bge 0xfeaaaaa0
fffffff4 aaaaaaaa bge 0xfeaaaaa4
fffffff8 aaaaaaaa bge 0xfeaaaaa8
fffffffc aaaaaaaa bge 0xfeaaaaac
1 .global _start
2 _start:
3 ldr r0,=0xff200000
_start:
00000000 e59f0048 ldr r0, [pc, #72] (0x50: end)
4 ldr r1,=0xff200020
00000004 e59f1048 ldr r1, [pc, #72] ; 0x54
5 ldr r8,=0xff200040
00000008 e59f8048 ldr r8, [pc, #72] ; 0x58
0000000c e598b000 6 ldr r11, [r8]
00000010 e598c000 7 ldr r12, [r8]
00000014 e00a0c9b 8 mul r10, r11, r12
00000018 e1a0500a 9 mov r5, r10
0000001c e2056001 10 and r6, r5, #1 ; 0x1
00000020 e3560001 11 cmp r6, #1 ; 0x1
Editor (Ctrl-E)
Disassembly (Ctrl-D)
Memory (Ctrl-M)
Messages
Compiling...
Code and data loaded from ELF executable into memory. Total size is 96 bytes.
Assemble: arm-eabi-as -mfloat-abi=softfp -march=armv7-a -mcpu=cortex-a9 -mfpu=neon-fp16 --gdwarf2 -o work/asmR2tav4.s.o work/asmR2tav4.s
Link: arm-eabi-ld --script build_arm.ld -e _start -u _start -o work/asmR2tav4.s.elf work/asmR2tav4.s.o
Compile succeeded.
Clear
Devices
LEDs ff200000
Switches ff200040
Push buttons IRQ 73 ff200050
Seven-segment displays ff200020
JTAG UART IRQ 80 ff201000
Read FIFO: 0
Write FIFO: 0
Cortex-A9 Private Timer IRQ 29 fffec600
0 Once Stop TO=0
Cortex-A9 Watchdog Timer IRQ 30 fffec620
0 Once Stop TO=0 RST=0
HPS L4 Watchdog Timer IRQ 203 ffd02000
2147483648 Interrupt Stop INTR=0
HPS L4 Watchdog Timer IRQ 204 ffd03000
2147483648 Interrupt Stop INTR=0
Search
9:24 AM
7/12/2025


Stopped
Step Into F2
Step Over Ctrl-F2
Step Out Shift-F2
Continue F3
Stop F4
Restart Ctrl-R
Reload Ctrl-Shift-L
File
Help
Registers
Refresh
r0 ff200000
r1 ff200020
r2 00000000
r3 00000000
r4 00000000
r5 00000000
r6 00000000
r7 00000000
r8 ff200040
r9 00000000
r10 00000000
r11 00000005
r12 00000006
sp 00000000
lr 00000000
pc 00000014
cpsr 000001d3 NZCVI SVC
spsr 00000000 NZCVI ?
Registers
Call stack
Trace
Breakpoints
Watchpoints
Symbols
Counters
Settings
Number Display Options
Size: Word
Format: Hexadecimal
Memory words per row: 4
Disassembly (Ctrl-D)
Go to address, label, or register: 00000000
Refresh
Address Opcode Disassembly
ffffffd8 aaaaaaaa bge 0xfeaaaa88
ffffffdc aaaaaaaa bge 0xfeaaaa8c
ffffffe0 aaaaaaaa bge 0xfeaaaa90
ffffffe4 aaaaaaaa bge 0xfeaaaa94
ffffffe8 aaaaaaaa bge 0xfeaaaa98
ffffffec aaaaaaaa bge 0xfeaaaa9c
fffffff0 aaaaaaaa bge 0xfeaaaaa0
fffffff4 aaaaaaaa bge 0xfeaaaaa4
fffffff8 aaaaaaaa bge 0xfeaaaaa8
fffffffc aaaaaaaa bge 0xfeaaaaac
1 .global _start
2 _start:
3 ldr r0,=0xff200000
_start:
00000000 e59f0048 ldr r0, [pc, #72] (0x50: end)
4 ldr r1,=0xff200020
00000004 e59f1048 ldr r1, [pc, #72] ; 0x54
5 ldr r8,=0xff200040
00000008 e59f8048 ldr r8, [pc, #72] ; 0x58
0000000c e598b000 6 ldr r11, [r8]
00000010 e598c000 7 ldr r12, [r8]
00000014 e00a0c9b 8 mul r10, r11, r12
00000018 e1a0500a 9 mov r5, r10
0000001c e2056001 10 and r6, r5, #1 ; 0x1
00000020 e3560001 11 cmp r6, #1 ; 0x1
12 beq odd
Editor (Ctrl-E)
Disassembly (Ctrl-D)
Memory (Ctrl-M)
Messages
Compiling...
Code and data loaded from ELF executable into memory. Total size is 96 bytes.
Assemble: arm-eabi-as -mfloat-abi=softfp -march=armv7-a -mcpu=cortex-a9 -mfpu=neon-fp16 --gdwarf2 -o work/asmR2tav4.s.o work/asmR2tav4.s
Link: arm-eabi-ld --script build_arm.ld -e _start -u _start -o work/asmR2tav4.s.elf work/asmR2tav4.s.o
Compile succeeded.
Clear
Devices
LEDs ff200000
Switches ff200040
Push buttons IRQ 73 ff200050
Seven-segment displays ff200020
JTAG UART IRQ 80 ff201000
Read FIFO: 0
Write FIFO: 0
Cortex-A9 Private Timer IRQ 29 fffec600
0 Once Stop TO=0
Cortex-A9 Watchdog Timer IRQ 30 fffec620
0 Once Stop TO=0 RST=0
HPS L4 Watchdog Timer IRQ 203 ffd02000
2147483648 Interrupt Stop INTR=0
HPS L4 Watchdog Timer IRQ 204 ffd03000
2147483648 Interrupt Stop INTR=0
Search
9:24 AM
7/12/2025


Lab 5 Task.png: CSE331L8 Micr
MD. Rokib Hasan Oli 2211950642-6
CPUlator ARMv7 System Simulator
CPUlator ARMv7 System Simulator
Binary Calculator
Private browsing
cpulator.01xz.net/?sys=arm-de1soc&d_audio=48000
Import bookmarks...
Finish setup
Stopped
Step Into F2
Step Over Ctrl-F2
Step Out Shift-F2
Continue F3
Stop F4
Restart Ctrl-R
Reload Ctrl-Shift-L
File
Help
Registers
Refresh
r0 ff200000
r1 ff200020
r2 00000000
r3 00000005
r4 00000077
r5 0000001e
r6 00000000
r7 00000000
r8 ff200040
r9 00000000
r10 0000001e
r11 00000005
r12 00000006
sp 00000000
lr 00000000
pc 00000050
cpsr 800001d3 NZCVI SVC
spsr 00000000 NZCVI ?
Registers
Call stack
Trace
Breakpoints
Watchpoints
Symbols
Counters
Settings
Number Display Options
Size: Word
Format: Hexadecimal
Memory words per row: 4
Disassembly (Ctrl-D)
Go to address, label, or register: 00000000
Refresh
Address Opcode Disassembly
00000028 e3a03005 mov r3, #5 ; 0x5
0000002c e5803000 str r3, [r0]
00000030 e3a04077 mov r4, #119 ; 0x77
00000034 e5814000 str r4, [r1]
00000038 ea000004 b 0x50 (0x50: end)
odd:
0000003c e3a03002 mov r3, #2 ; 0x2
00000040 e5803000 str r3, [r0]
00000044 e3a0407c mov r4, #124 ; 0x7c
00000048 e5814000 str r4, [r1]
0000004c eaffffff b 0x50 (0x50: end)
end:
00000050 ff200000 svc #2097152 ; 0x200000
00000054 ff200020 svc #2097184 ; 0x200020
00000058 ff200040 svc #2097216 ; 0x200040
Editor (Ctrl-E)
Disassembly (Ctrl-D)
Memory (Ctrl-M)
Messages
Clear
Compiling...
Code and data loaded from ELF executable into memory. Total size is 96 bytes.
Assemble: arm-eabi-as -mfloat-abi=softfp -march=armv7-a -mcpu=cortex-a9 -mfpu=neon-fp16 --gdwarf2 -o work/asmR2tav4.s.o work/asmR2tav4.s
Link: arm-eabi-ld --script build_arm.ld -e _start -u _start -o work/asmR2tav4.s.elf work/asmR2tav4.s.o
Compile succeeded.
Devices
LEDs ff200000
Switches ff200040
Push buttons IRQ 73 ff200050
Seven-segment displays ff200020
JTAG UART IRQ 80 ff201000
Read FIFO: 0
Write FIFO: 0
Cortex-A9 Private Timer IRQ 29 fffec600
0 Once Stop TO=0
Cortex-A9 Watchdog Timer IRQ 30 fffec620
0 Once Stop TO=0 RST=0
HPS L4 Watchdog Timer IRQ 203 ffd02000
2147483648 Interrupt Stop INTR=0
HPS L4 Watchdog Timer IRQ 204 ffd03000
2147483648 Interrupt Stop INTR=0
Search
9:24 AM
7/12/2025